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A novel approach to diagnosing PCOS through tunica albuginea oculi

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ABSTRACT: Polycystic ovarian syndrome (PCOS) mostly affects women aged fifteen to forty-four years. Using patient tunica albuginea oculi images, we created a deep learning model in this study to predict PCOS. In individuals with PCOS, physiological alterations have been observed in the tunica albuginea oculi, a highly vascularized tissue. Our model's focus on the relevant traits present in the tunica albuginea oculi was strengthened by employing a strong image segmentation technique to separate the tunica albuginea oculi region from the entire eye image. Following segmentation, we classified the tunica albuginea oculi images as healthy or PCOS using a pre-trained deep learning model called Squeeze-and-Excitation Networks (SENet). In terms of prediction, the suggested system performed quite well.

Keywords: Polycystic ovary syndrome, PCOS, tunica albuginea oculi images, non-invasive diagnosis, deep learning, image segmentation, SENet

1 INTRODUCTION

1.1 *PCOS and its significance*

Polycystic ovarian syndrome (PCOS) is one of the most widespread hormonal diseases in women across the globe. It is assumed that the percentage is within 6%–26%. Mainly, it is higher in Indian women with rates varying between 3.7% and 22.5%, and it is the most hormonal problem among this group. Several factors lead to the occurrence of PCOS, these include lack of workout, lifestyle, genes, and unhealthy eating habits that result in obesity thereby disrupting hormone function.

1.2 *Motivation and background*

At our educational institution, we started research to understand the difficulties people with PCOS face. The incidence of PCOS is 90 out of 256. Moreover, we uncovered that a primary concern for nearly all patients in remote and rural medical facilities was the complexity of taking medicine on time. They also face expensive, time-consuming travel to cities for medical services. This information aims to the objective of solving some of the challenges that people with PCOS experience.

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1.3 *Current available medication*

PCOS is diagnosed using ultrasound imaging, clinical assessment, and blood tests to detect polycystic ovaries, raised testosterone, and irregular menstrual cycles. Ovarian cysts are visualized with ultrasound scans while hormone levels are determined using blood tests. These techniques are expensive and resource-intensive, hence not accessible for the less privileged or those in remote places. This complexity and time-consuming nature of these diagnostics.

1.4 *Advanced approach wants to be introduced*

Compared to traditional detection methods, our approach is more economical and reachable, allowing early identification of PCOS. By using innovative technologies, we address the time-consuming and costly nature of current procedures, offering hope for those in remote areas or with limited financial means. The intuitive interface promotes awareness and proactive health management. Our initiative advances PCOS public health outcomes, diagnostic technology, and healthcare accessibility.

2 LITERATURE REVIEW

There are ovarian cysts that lead to numerous challenges including weight gain, skin disorders as well as difficulty in conceiving some women. Early recognition is very essential since it helps in preventing the worsening of the symptoms and providing of right cure for it. Specialized computer models within the existing data have succeeded in classifying the cysts with very high accuracy. Image improvement, size analysis, and computer patterning have been suggested using ultrasound scans to obtain detailed information regarding ovaries [1]. Polycystic Ovary Syndrome (PCOS), is a major international health issue that may lead to long-term effects such as type 2 diabetes mellitus or gestational diabetes when early detected and treated appropriately. There is hope that machine learning methods can be effective tools in disease diagnosis because they exploit data to advance comprehension and fine-tune model performance. The ensemble classifier managed to deliver extraordinary accuracy rates when evaluated using featured subsets while maintaining consistent performance [4]. Polycystic ovary syndrome is a common hormone disorder that causes problems with getting pregnant among women. Metabolic problems, hormonal imbalances, and immune system confusion are the causes of this problem. This study will look at how these two important systems work together towards causing autoimmune diseases in PCOS patients [6]. A significant health challenge, which involves around 10 percent of women, leads to irregular periods, fertility problems, and excess hair growth because of high hormone levels. This is called Polycystic ovary syndrome (PCOS) [7]. A study in Sudan was conducted on the medical and chemical characteristics of different types of endometrial polyps. The study, which found that oligo/anovulation was the most common type of endometrial polyps among Sudanese women with PCOS, suggested that more epidemiological studies should be carried out based on geographical, ethnic, and genetic features [8]. Polycystic Ovary Syndrome (PCOS) afflicts many women. It is defined by metabolic imbalances manifesting in irregular menstruation, infertility problems, and hypertrichosis caused by disrupted hormones; invariably these illnesses may lead to metabolic problems like insulin resistance and abnormal blood fats [9]. PCOS was detected in a group of five hundred and forty-one individuals using machine learning methods, specifically linear support vector machine which proved to distinguish cases of PCOS from the rest of the dataset thereby illustrating how important machine learning is in improving accuracy and making timely diagnosis [10]. Research conducted by AIIMS, Patna says that people experience the symptoms of PCOS such as late menstruation and miscarriage resulting in a lack of good living conditions for the women [11]. It was found that when health and mortality were compared between women older than 80 years old who had polycystic ovary syndrome (PCOS) and those of similar number but no PCOS there was nothing significant concerning mortality rates or incidences of new heart disease during 12 years. In conclusion, the study indicates that such risk factors as type two diabetes do not lead to women developing an additional risk factor for coronary illness associated with PCOS [12]. Women in the age group of 25–35 are affected by PCOS, with a "Furious Flies" system that uses a classifier and artificial neural network to recognize features at an accuracy level of 98.63% that are detected than other methods [13]. They used the Python Scikit Learn tool and RapidMiner to foresee PCOS, during the process RapidMiner excelled over Python [14].

3 FRAMEWORK DEVELOPMENT

3.1 *Methodology*

We present a method of predicting PCOS with the help of the SE networks model based on photos of fabric oculi and the squeeze-and-excitation (SE) networks model (Figure 1). Using the SE model's ability to capture intricate feature relationships, we manage to pull out key characteristics from preprocessed pictures. A classification algorithm was created using these characteristics. The large dataset for training and validation purposes has been analyzed in detail, with such methods as data augmentation and ensemble used in such analysis.

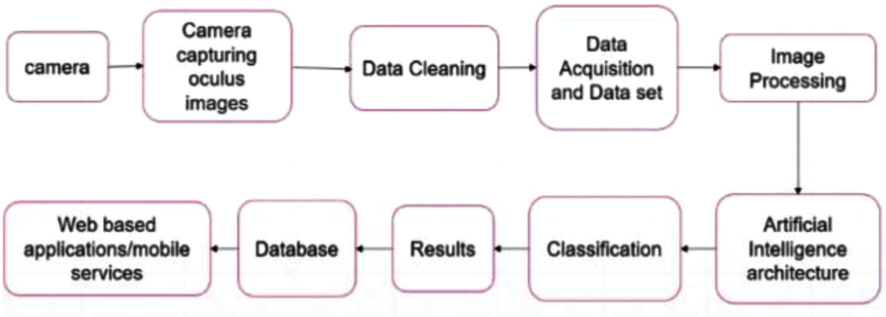

Figure 1. Methodology.

3.2 *Data collection and preparation*

We aimed to create an eye image-based polycystic ovary syndrome forecasting model through a project, using a dataset that is both representative and diverse. Randomly selected women from the university campus decided whether they had any indicators of PCOS or not then they were categorized into those who had the symptoms of PCOS as well as the ones who did not have it so that we would have an equal number in each category. Figure 2 shows the input image, Figure 3 shows the segmented image and Figure 4 shows the blood vessels that are highlighted.

Figure 2. Input data.

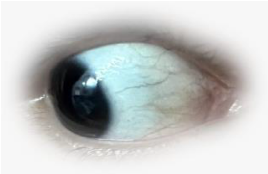

Figure 3. Segmented image.

Figure 4. Highlighting blood vessels.

3.3 *Image collection, image preprocessing and image segmentation*

Using a high-resolution camera in a controlled setting, we took six pictures from both eyes of participants. This was done to ensure uniform accurate results all around. Through this process, we were able to get well-detailed images of the tunica albuginea oculi for training purposes. So we had a data set of detailed images of the tunica albuginea oculi from the eyes of the participants. We extracted the tunica albuginea oculi region from the eye images using a robust segmentation technique, enabling our model to focus on relevant features and improve accuracy.

```
print(classification_report(y_test, y_pred, target_names=['no','yes']))

              precision    recall  f1-score   support

          no       0.92      0.85      0.88        13
         yes       0.85      0.92      0.88        12

    accuracy                           0.88        25
   macro avg       0.88      0.88      0.88        25
weighted avg       0.88      0.88      0.88        25
```

Figure 5. Classification report of PCOS prediction model.

4 RESULTS

The results reveal significant enhancement in the PCOS prediction efficiency by the use of tunica albuginea oculi images and SE networks as shown in Figure 5. Our model attained an 88% general accuracy following data augmentation and advanced training techniques. The predictability of the ‘no’ class is at a 92% precision while it stands at 85% in the ‘yes’ category with recall rates of 85% and 92%. The fact that a balanced F1–score for both classes is 88% shows its robust performance in effectively predicting this condition.

5 CONCLUSIONS

This study aims to use sophisticated deep-learning models to predict PCOS accurately by identifying eye patterns among persons who manage Polycystic Ovary Syndrome (PCOS) in rural areas. The main objectives of this project comprise the promotion of earlier diagnosis; and reduction of healthcare services provision barriers while encouraging increased self-care support among the patients thereby enhancing exceptional healthcare solutions in the end.

6 FUTURE SCOPE

We have developed cutting-edge deep learning models, like the Squeeze-and-Excitation Network, that accurately predict PCOS in our project. Our objective is to provide new meanings for diagnosing PCOS, the future possibility of model improvement, integrating multimodal data, and developing mobile applications for users' convenience. The proactive management of PCOS involves monitoring in real-time and global cooperation to provide a healthcare experience that will change lives. We are the people who enable others while at the same time connecting them through health care.

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